



«Approved»
by the Dean
Syzdykova Z.A.
«___» of _____ 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	Big Data Methods and Tools
Degree cycle (level)/major	Master (ADA)
Year, term	Second year, 1 st term
Number of credits	5
Language of delivery:	English, Russian
Prerequisites	Statistics, Programming
Postrequisites	Research methods
Lecturer(s)	Oleksandr Kuchanskyi, D.Sc. in Information Technology, Professor, a.kuchanskyi@astanait.edu.kz , Astana IT University, Expo, C1 block, office C1.2.331.
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	The course focuses on developing computational research skills, concentrating on big data mathematical methods and tools. It will analyze the major trends driving the rise of big data and data science problems, discuss the principal component analysis for the dimension reduction of massive data sets and discuss efficient algorithms, the hardness of problems involving various types of big data
2. Course Goal(s)	The course aims to provide master students with knowledge of big data mathematical methods, data science problems, and tools to solve applied problems.
3. Course Objectives:	The course objectives are: • provide information about big data mathematical methods, data science problems; • provide information on efficient algorithms, the hardness of problems involving various types of big data
4. Skills & Competences	The skills covered in the course are: • skill to understand how to solve big data and data science problems; • programming skills.
5. Course Learning Outcomes:	By the end of this course master students will be able: • to define basic concepts of big data. • to understand basic mathematical methods for data science. • to evaluate, interpret and report the results. • to produce new value from available data.

6. Methods of Assessment	Methods of assessment include: • Two midterm quizzes. • Four practical assignments.
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1. Chen, Li. M., Su, Z., Bo, J. (2015). Mathematical Problems in Data Science: Theoretical and Practical Methods. Springer. 219 p. 2. Phillips, J.M. (2018). Mathematical Foundations for Data Analysis. 175 p. 3. Peyre, G. (2021). Mathematical Foundations of Data Science. CNRS&DMA. 294 p. 4. Breiding, P., Fairchild, S. (2022). Mathematical Methods in Data Science. Leipzig. 117 p. 5. Ren, J., Wang, H. (2023). Mathematical Methods in Data Science. Elsevier. 246 p. <u>Supplementary literature:</u> 6. Roberts, D. A., Sho Yaida, Boris H. (2022). The Principles of Deep Learning Theory. An Effective Theory Approach to Understanding Neural Networks 7. Prince, S. J.D. (2023). Understanding Deep Learning. 544 p.
8. Resources	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
9. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material

provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing.

Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties:

- Offers a different and unique, but relevant, perspective;
- Contributes to moving the discussion and analysis forward;
- Builds on other comments.

Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes.

Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor.

Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Final exam* / Final project:

The final exam is conducted in the form of a test.

	Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
PART I. MATHEMATICAL PROBLEMS IN DATA SCIENCE						
1	Topic 1. Introduction: Data Science and Big Data Computing - Problems related to Big Data: - Data Science in Industry.	3	2	0	1	9
2	Topic 2. Overview of Basic Methods for Big Data (part 1) «Hardware» and «Software» of Data Science: - Probability review, convergence and sampling.	3	2	0	1	9
3	Topic 3. Overview of Basic Methods for Big Data (part 2) - Distances and nearest neighbors: - Linear regression and gradient descent.	3	2	0	1	9
4	Topic 4. Overview of Basic Methods for Big Data (part 3) - Principal component analysis: - Linear Algebra review.	3	2	0	1	9
5	Topic 5. Overview of Basic Methods for Big Data (part 4) - Clustering and Classification: - Graph theory.	3	2	0	1	9
PART II. DATA SCIENCE PROBLEMS AND SELECTED TOPICS IN BIG DATA						
6	Topic 6. Statistical Learning Algorithms and Big Data Decision Tree, Neural Network, Genetic Algorithms, SVM.	3	2	0	1	9
7	Topic 7. Time Series Data Analysis - Nonlinear dynamics and chaos theory: - Example of TSDA in Data Science	3	2	0	1	9
8	Topic 8. Images, Videos and Big Data Big data related image segmentation.	3	2	0	1	9

9	Topic 9. Monte Carlo Methods in Big Data Analysis - Variance reduction; - Example of Monte Carlo Methods in Data Science.	3	2	0	1	9
10	Topic 10. Selected Topics in Big Data Advances methods in Big Data, its applications and tools.	3	2	0	1	9
	Total hours: 150	30	20	0	10	90

3.2 List of Assignments for Student Independent Study

Nº	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
PART I. MATHEMATICAL PROBLEMS IN DATA SCIENCE				
1	Practical Task -1	9	Basic literature [1 – 7]	A report
2	Practical Task -2	9	Basic literature [1 – 7]	A report
PART II. DATA SCIENCE PROBLEMS AND SELECTED TOPICS IN BIG DATA				
3	Practical Task -3	9	Basic literature [1 – 7]	A report.
4	Practical Task -4	9	Basic literature [1 – 7]	A report.

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignments: - Assignment 1. - Assignment 2. Mid Term	60 30 30 40	100
2 nd attestation	Assignments: - Assignment 3. - Assignment 4. End Term	60 30 30 40	100
Final exam*	Final Test	100	100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

**** The number of assignments can be different. It depends from the course program and designed by**

course syllabus. But the total points of the assignment are 60 in each control period.

- * Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4.0	95-100	Excellent
A-	3.67	90-94	
B+	3.33	85-89	Good
B	3.0	80-84	
B-	2.67	75-79	
C+	2.33	70-74	
C	2.0	65-69	Satisfactory
C-	1.67	60-64	
D-	1.33	55-59	
D	1.0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete

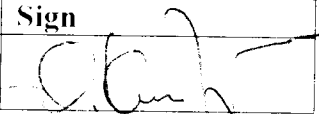
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

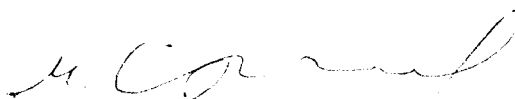
Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Oleksandr Kuchanskyi	Professor	26.08.2024	

Director



Sergaziyev M.Zh.